Comprehensive Codebase Architecture & Empirical Validation Report

FLARE-CHRONOS

Codebase Architecture, Deep-Module Engineering, Empirical Validation & Dual-System Cognitive Harness Report

Challenge: *NASA Space Apps Challenge 2026 — Harmonization of MODIS and VIIRS Hot Spots*
Document Version: `v1.1.0` · **Test Suite:** `182 / 182` Offline Tests + `6 / 6` Live Tests (`22` Test Files) · **AI Benchmark:** `15 / 15` Use Cases (`100%`)

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1. Executive Summary & Key Engineering Achievements

This technical report documents the software architecture, physical equations, empirical data provenance, cognitive AI orchestration, interactive web workbench, and automated verification suite of **FLARE-CHRONOS**.

Engineering Dimension	Verified Implementation & Empirical Outcome
1. Deep-Module Software Architecture	Strictly isolates pure physical/mathematical computation (`packages/harmonization-core` , **zero I/O**) from network/ disk adapters (`packages/firms-ingestion`), experiment runners (`apps/benchmark-cli`), cognitive orchestration (`apps/ai-service` , port `5901`), and the interactive web surface (`apps/workbench` , port `5900`).
2. Real NASA FIRMS Archive Provenance	Commits `134` **non-empty NASA FIRMS CSV slices (`1,609,182` total detections; `451,488` detections in the canonical 5-day peak windows)** pinned by `data/raw/SHA256SUMS` and `data/raw/fetch_manifest.json` , enabling 100% offline reproducibility after `git clone` .

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Engineering Dimension	Verified Implementation & Empirical Outcome
3. Cross-Sensor Radiometric Calibration	Fits a two-stage scan-angle + Stratified Empirical Quantile Mapping (EQM) transfer operator across **N = 20,786 co-incident Aqua–SNPP grid cells**, reducing Kolmogorov-Smirnov distance $\mathbf{D_{KS}}$ from `0.1153` to `0.0353` (`-69.4%`) and raising log-FRP $\mathbf{R^2}$ from `0.7839` to `0.8048` .
4. Multi-Sensor Step-Jump Elimination	Removes the `+1,311.2%` **(`+1.311,2%`) artificial step-jump** in the Mekong Subregion (`2024` raw index `1,411.2` → harmonized index `90.5` , `-9.5%` vs. `2005 = 100.0`) and compresses `112,405` raw Amazon `2024` detections `5.75×` into `19,542` independent `FireEpisodes` (harmonized index `37.7`).
5. 26-Year Burning Activity Calendar (`26 × 73`)	Computes Harmonized Fire Density (**HFD**), Harmonized Radiative Flux (**HRF**), and Standardized Fire Anomaly Index (**SFAI** Z-score) across **25 Measured Epochs (`2001–2025`)** and **1 Interpolated Year (`2026`)** per region (ADR-0009, ADR-0010).
6. Dual-System Cognitive AI Harness (`apps/ai-service`)	Orchestrates an 8-node LangGraph `StateGraph` across `15` **Use Cases (`5` System 1 + `10` System 2)** and `18` **registered model-facing `@tool` functions (`9` Atomic + `9` Domain Diagnostic, `19` total with internal CSV tool)**, achieving `15/15` **(`100%`)** pass rate in both offline deterministic and live Gemini benchmarks with `0` unverified numeric claims.
7. Automated Quality & Consistency Guardrails	Enforced by `182` **offline pytest tests + `6` live TDD seam tests (`188` total across `22` test files in `tests/`)**, `ruff` , `pyright` , OpenAPI schema drift checks, and `scripts/check-docs.py` .

2. Five Physical Discontinuities Addressed by the Codebase

#	Discontinuity	Physical Mechanism	Responsible Module & Function
1	**Bow-Tie Scan Footprint Expansion**	MODIS pixel grows **9.69 ×** (**1.0000 → 9.6875 km²**) at **θ = 55°**; **74.23 ×** larger than VIIRS nadir (**0.1305 km²**). Empirical FIRMS mean: `2.17 km²` (P95 `5.27 km²`) for MODIS vs. `0.22 km²` (P95 `0.42 km²`) for VIIRS (`9.86×` ratio).	`geometry_h3.py` : `invert_modis_scan_angle` , `viirs_dimensions_from_scan_angle` , `allocate_footprint_to_h3`
2	**Radiometric $FRP_{lim}(\theta)$ Blindness**	MODIS daytime detection limit reaches **41.20 MW** at scan edge vs. VIIRS **3.40 MW** (**12.12 ×** gap).	`radiometric_calibrator.py` : `modis_minimum_detectable_frp` , `fit_coincident_aqua_snpp_model`
3	**Afternoon Overpass Double-Counting**	`Aqua` , `S-NPP` , `NOAA-20` , and `NOAA-21` cross the same fire within 25–95 minutes (`13:30 / 01:30 MLT`).	`episode_clustering.py` : `cluster_hotspots_into_episodes` , `compute_effective_observation_effort`

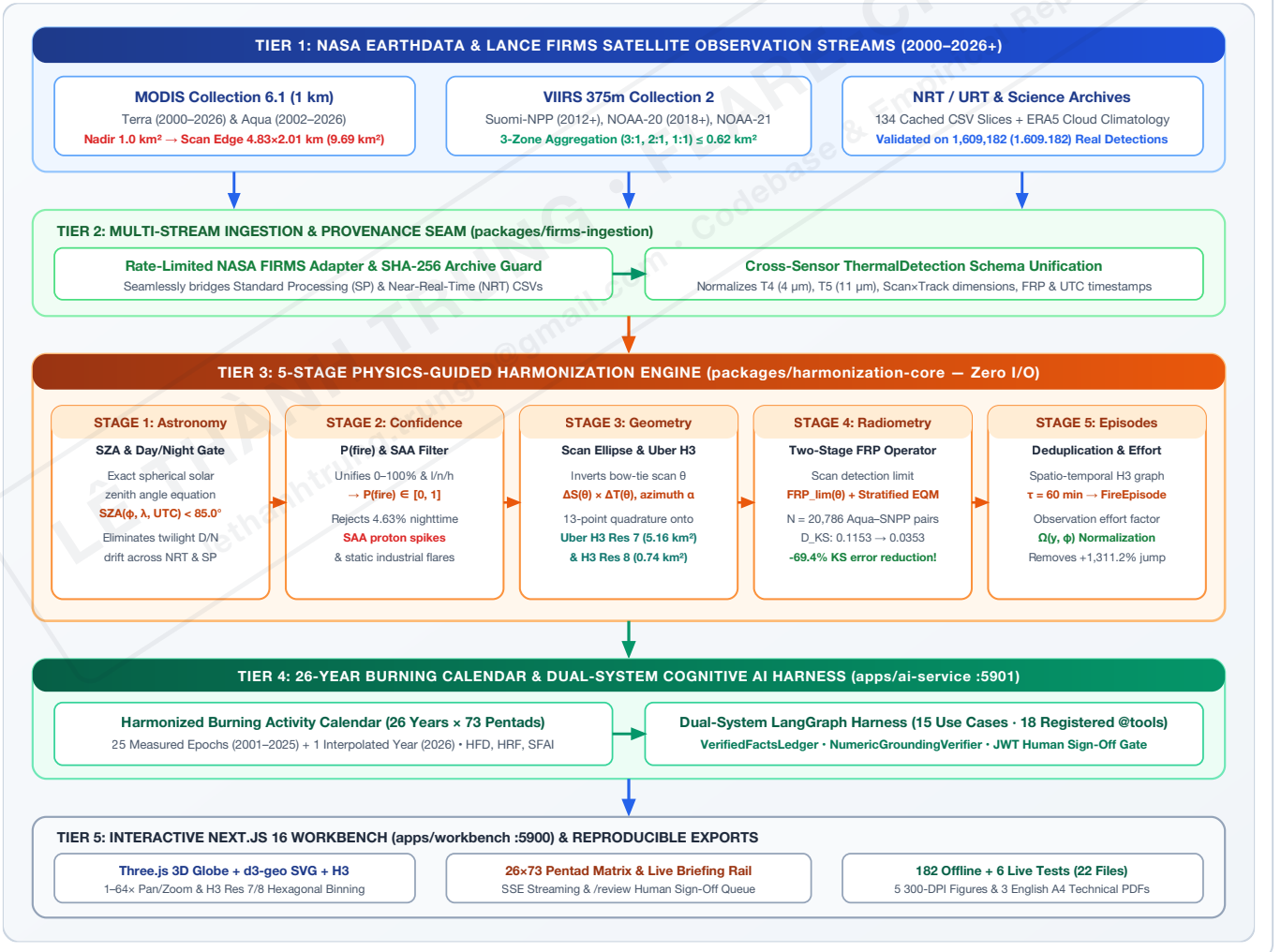
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#	Discontinuity	Physical Mechanism	Responsible Module & Function
4	**Day/Night Flag Drift (`URT/RT/NRT` vs. `SP`)**	Twilight detections (**SZA ≈ 85°**) carry inconsistent `D/N` flags between NRT and Standard streams.	`astronomy_sza.py` : `compute_solar_zenith_angle` , `unify_daynight_flag`
5	**South Atlantic Anomaly (SAA) & Static Flare Noise**	Cosmic-ray proton strikes contaminate `4.63%` of nighttime S-NPP detections over South America.	`confidence_filter.py` : `is_south_atlantic_anomaly_spike` , `filter_and_score_detections`

3. Deep-Module Monorepo Architecture (`packages/` , `apps/` , `src/` , `tests/`)

3.1. End-to-End Monorepo Component Architecture

ARCHITECTURE FIGURE 1: 5-TIER END-TO-END FLARE-CHRONOS CROSS-SENSOR HARMONIZATION & COGNITIVE WORKBENCH ARCHITECTURE



3.2. Complete Repository Tree

```
flare-chronos/
├── GLOSSARY.md / GLOSSARY-MAP.md          # Multi-context ubiquitous domain language
├── DESIGN.md                              # Workbench visual design system, tokens & EN/VI copy
├── pyproject.toml                         # uv Python workspace root
├── packages/
│   ├── harmonization-core/                # [Deep Module] Pure physics & statistics (zero I/O)
│   │   └── src/flare_chronos_core/
│   │       ├── schemas.py                 # Immutable domain dataclasses & SFAI constants
│   │       ├── astronomy_sza.py           # Exact spherical Solar Zenith Angle (< 85°)
│   │       ├── confidence_filter.py       # Unified P(fire), SAA cosmic-ray & static source filters
│   │       ├── geometry_h3.py             # Bow-tie scan inversion & 13-point H3 Res 7/8 quadrature
│   │       ├── radiometric_calibrator.py  # Two-stage FRP_lim(θ) + Stratified EQM calibrator
│   │       ├── episode_clustering.py      # O(N) H3 spatio-temporal deduplication & effort Ω(y,φ)
│   │       ├── burning_calendar.py        # 26 Years × 73 Pentads HFD / HRF / SFAI engine
│   │       └── pipeline.py                # End-to-end pure functional pipeline orchestrator
│   └── firms-ingestion/                   # [Adapter Seam] NASA LANCE FIRMS HTTP client + CSV parser
│       └── src/firms_ingestion/
│           ├── firms_client.py            # Rate-limited FIRMS API client & local slice cache
│           └── csv_loader.py              # Schema-normalizing MODIS/VIIRS CSV loader
├── apps/
│   ├── benchmark-cli/                     # [Application] Empirical runner & 300-DPI figure renderer
│   ├── ai-service/                        # [Cognitive Layer] Dual-System LangGraph + FastAPI (:5901)
│   │   └── src/ai_service/
│   │       ├── schemas.py                 # 15 UseCaseIds, VerifiedFactsLedger, GroundedFireBriefing
│   │       ├── tools.py                   # 18 registered @tools (9 Atomic + 9 Diagnostic) + 1 CSV tool
│   │       ├── models.py                  # Explicit EndpointProfiles & Grounded Narrative synthesizer
│   │       ├── harness.py                 # 8-node LangGraph Dual-System StateGraph
│   │       ├── api.py / serve.py          # FastAPI ASGI server (:5901), JWT auth, SSE & sign-off
│   │       └── store.py / review.py       # Append-only SQLite RunStore & review queue projection
│   └── workbench/                         # [Web Surface] Next.js 16 + Three.js + d3-geo SVG + H3 (:5900)
├── exports/
│   ├── PROPOSAL.md / PROPOSAL.pdf         # Official Technical Proposal (English A4 PDF)
│   ├── REPORT.md / REPORT.pdf             # Codebase Architecture & Empirical Report (English A4 PDF)
│   ├── MODEL_FLOW.md / MODEL_FLOW.pdf     # Dual-System Cognitive Model Flow Report (English A4 PDF)
│   └── export_pdf.py                      # Vector SVG + WeasyPrint A4 PDF compiler
├── scripts/
│   ├── canonical_facts.py                 # Single-source-of-truth metrics & claims registry
│   └── check-docs.py                      # Automated doc, version, and claim consistency scanner
└── tests/                                 # 182 offline pytest tests + 6 live tests across 22 files
```

4. Deep-Dive: `packages/harmonization-core` (Zero-I/O Physics Core)

1. `astronomy_sza.py`: Computes exact spherical Solar Zenith Angle **cos(SZA) = sinϕsinδ + cosϕcosδcos H** from UTC timestamp and coordinates, enforcing **SZA < 85.0°** as the unified daytime boundary across all `Standard` and `NRT` streams.
2. `confidence_filter.py`: Maps MODIS numeric confidence (`0–100%`) and VIIRS categorical confidence (`low` / `nominal` / `high`) onto a unified empirical probability **P(fire) ∈ [0, 1]**, rejects persistent industrial heat sources (**≥ 12** active pentads/year within a `750 m` radius), and removes nighttime single-pass proton spikes inside the South Atlantic Anomaly (`SAA`).
3. `geometry_h3.py`: Inverts MODIS `scan` (**Δ S**) to off-nadir angle **θ ∈ [0°, 55°]** via 45-iteration bisection (**< 10^{-9} rad** error), models VIIRS 3-zone hardware aggregation (`Zone 1: 3×1`, `Zone 2: 2×1`, `Zone 3: 1×1`), and integrates 13 quadrature points across each rotated footprint ellipse onto **Uber H3 Resolution 7 (`5.161 km²`)** and **Resolution 8 (`0.737 km²`)** cells with exact partition of unity (**$\sum w_{p,h} = 1.0$**).
4. `radiometric_calibrator.py`: Evaluates scan-angle minimum detectable **$FRP_{lim}(\theta)$** and applies the two-stage power-law + Stratified Empirical Quantile Mapping (EQM) transfer operator (**α = 1.7397, β = 0.9221, κ = 0.1000**).

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5. `episode_clustering.py` & `burning_calendar.py`: Deduplicates multi-satellite detections sharing a 1-ring H3 neighborhood within **τ = 60 min** into a single `FireEpisode` (fusing FRP by inverse variance), normalizes by effective observation effort **Ω(y, ϕ)**, and populates the `26 Years × 73 Pentads` matrix of **HFD**, **HRF**, and **SFAI** Z-scores.

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5. Deep-Dive: `packages/firms-ingestion` & `apps/benchmark-cli`

- **Archive Integrity (`data/raw/SHA256SUMS` & `fetch_manifest.json`)**: The repository caches `134` **non-empty NASA FIRMS CSV slices containing** `1,609,182` **total detections** (with `451,488` detections inside the canonical 5-day peak seasonal benchmark windows across Mekong and Amazon). `tests/test_raw_data_integrity.py` verifies every SHA-256 digest on every test run.
- **Automated Scientific Figure Generation (`figure_renderer.py`)**: Deterministically renders all five 300-DPI scientific figures (`fig1` through `fig5`) directly from `data/processed/empirical_results.json` and `data/processed/burning_calendar.json`.

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6. Empirical Validation & Scientific Figures (`1,609,182` Real NASA FIRMS Detections)

6.1. Summary of Empirical Validation Metrics

> **Note on Measured Epochs and Interpolated Years (ADR-0009, ADR-0010):**
> Both the Mekong (`2001–2026`) and Amazon (`2001–2026`) 26-year calendars consist of **25 Measured Epochs (`2001–2025`)** computed from real NASA FIRMS slices and **1 Interpolated Year (`2026`)** modelled from the in-progress MODIS count and regional seasonal envelope. Only Observed Cells in Measured Epochs (`n = 25` years) feed the SFAI baseline or receive an Anomaly Tier.

Study Region / Benchmark	Key Empirical Finding from `data/processed/empirical_results.json`
1. Mekong Subregion & Vietnam (`2001–2026`, `BBOX: 100.0°E–109.5°E, 8.5°N–23.5°N`)	Raw detections jump from `238` (`2001` Terra-only) and `3,914` (`2005` baseline = `100.0`) to `55,235` **in** `2024` **(Raw Index =** `1,411.2`**, an artificial step-jump of** `+1,311.2%` **/** `+1.311,2%`**)**. FLARE-CHRONOS compresses `2024` by **2.22 ×** to `24,902` episodes and normalizes by **Ω = 2.0645** to `8,150.0` **(`8.150,0`) VIIRS-eq episodes (Harmonized Index =** `90.5`**,** `−9.5%` **vs.** `2005`**)**, while preserving true El Niño peaks in `2007` (`237.9`, observed SFAI `+3.7σ`), `2010` (`184.9`), and `2016` (`133.0`).
2. Southern Amazon Basin (`2001–2026`, `BBOX: 62.0°W–50.0°W, 14.0°S–4.0°S`)	Filters `4.63%` nighttime SAA cosmic-ray spikes on S-NPP (`394 / 8,510` nighttime points). In `2024`, compresses `112,405` raw detections by `5.75×` (`5,75×`) into `19,542` `FireEpisodes` and normalizes by **Ω = 2.6716** to `4,942.4` **(`4.942,4`) VIIRS-eq episodes (Harmonized Index =** `37.7`**, raw index** `814.3`**)**. Observed SFAI peaks in `2002` at `+3.8σ` (`Pentad 47`).
3. Co-Incident Aqua–SNPP Calibration (N = 20,786 Pairs)	Across **N = 20,786** (`20.786`) co-incident grid cells, Kolmogorov-Smirnov distance D_{KS} drops from `0.1153` **to** `0.0353` **(`−69.4%`)**, log-FRP R^2 rises from `0.7839` **to** `0.8048`, and cell RMSE improves from `141.09` to `134.08 MW/cell`.

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6.2. Embedded Scientific Figures (`fig1` – `fig5`)

Figure 1 | Spherical Scan-Angle Footprint Physics & Dual-Resolution H3 Allocation
Comparison of MODIS C6.1 (1 km nominal) and VIIRS C2 (375 m 3-Zone Aggregated) Instantaneous Footprints + Real NASA FIRMS Observations

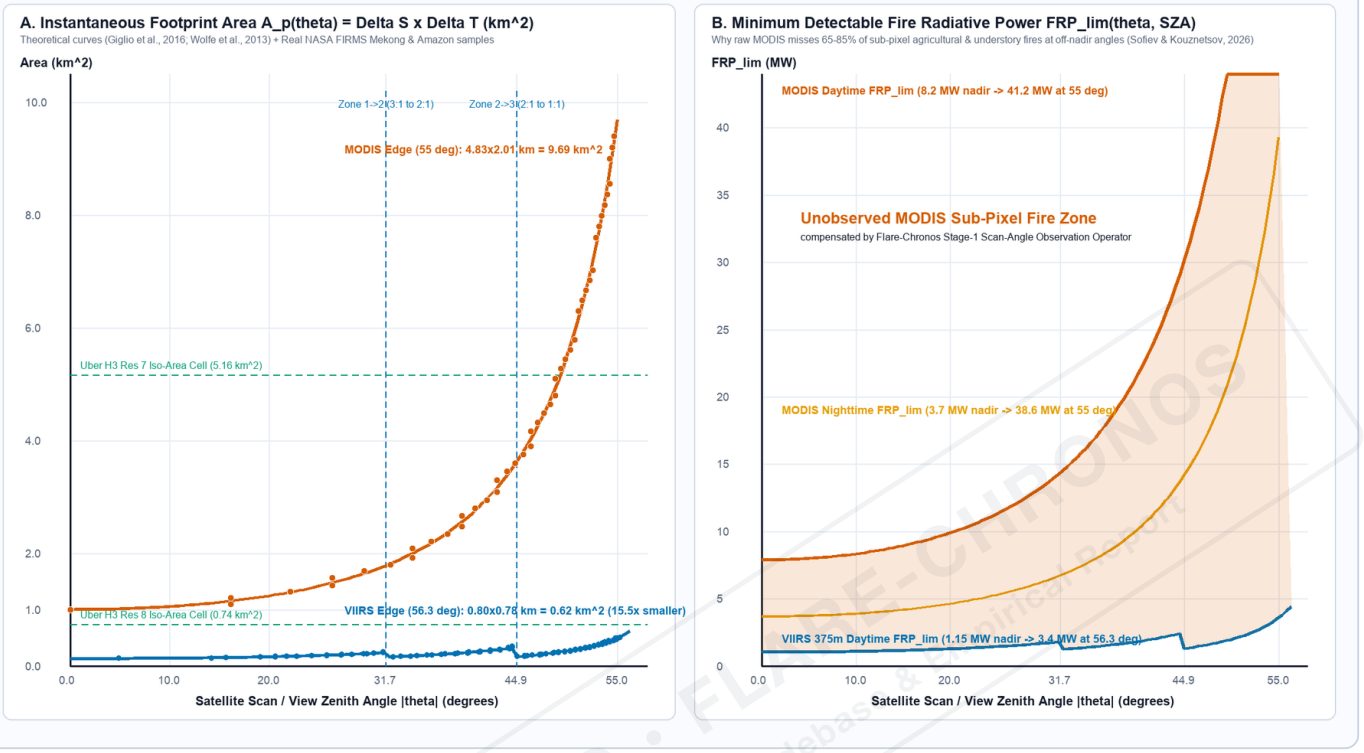


Figure 1 - Scan Angle Footprint Physics

Figure 2 | Spatio-Temporal Overpass Deduplication & Effective Observation Effort Omega(y, phi)
Resolving afternoon VIIRS constellation overlap (NOAA-20, NOAA-21, Suomi-NPP, Aqua) and NRT vs. Standard SZA discrepancies

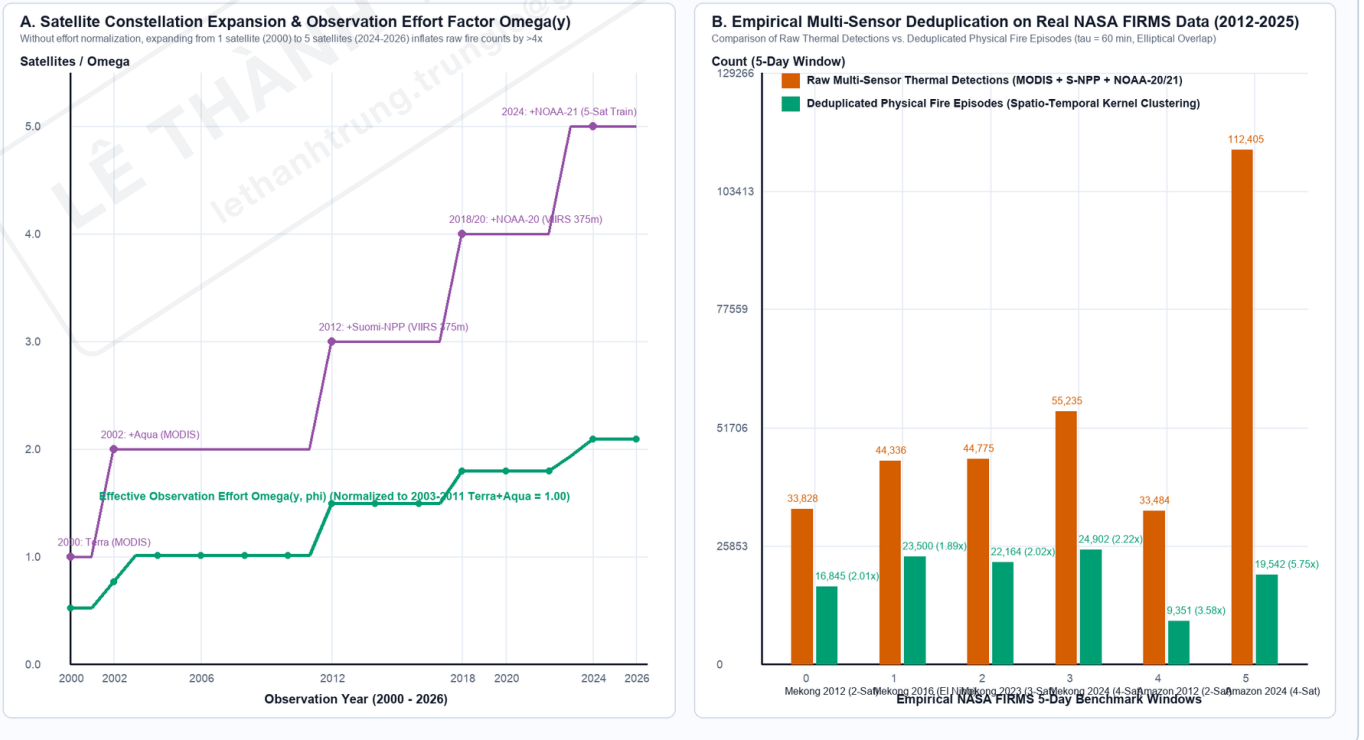


Figure 2 - Overpass Deduplication and Effort Factor

Figure 3 | Two-Stage Physics-Guided FRP Cross-Calibration (Co-Incident Aqua & Suomi-NPP Overpasses)

Stage 1 Scan-Angle Observation Operator + Stage 2 Empirical Quantile Mapping on N = 20786 Co-Incident Iso-Area Cells

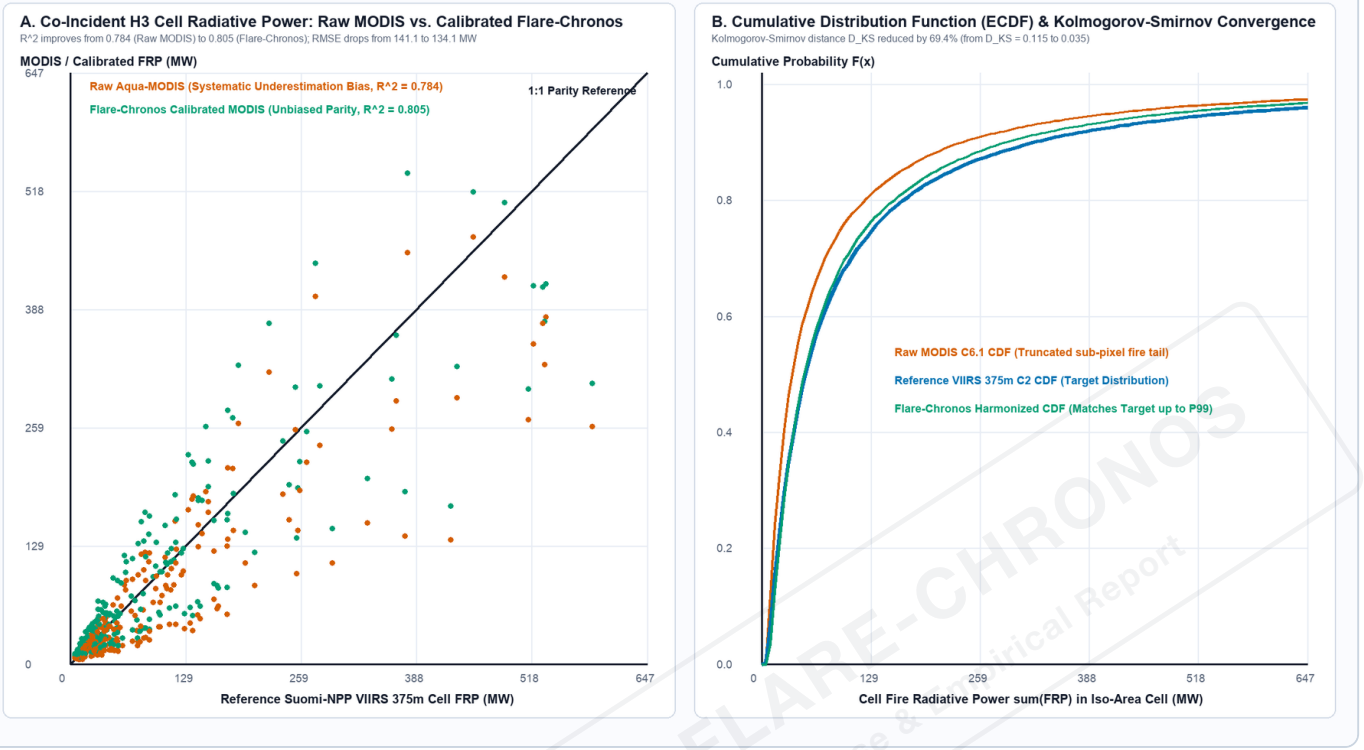


Figure 3 - Cross-Sensor FRP Calibration

Figure 4 | Elimination of Artificial Multi-Sensor Step Jumps Across 26 Years (2000 - 2026)

Demonstrating how raw unharmonized fire counts explode artificially in 2012 (+SNPP) and 2020/2024 (+NOAA-20/21) vs. Flare-Chronos Physical Baseline

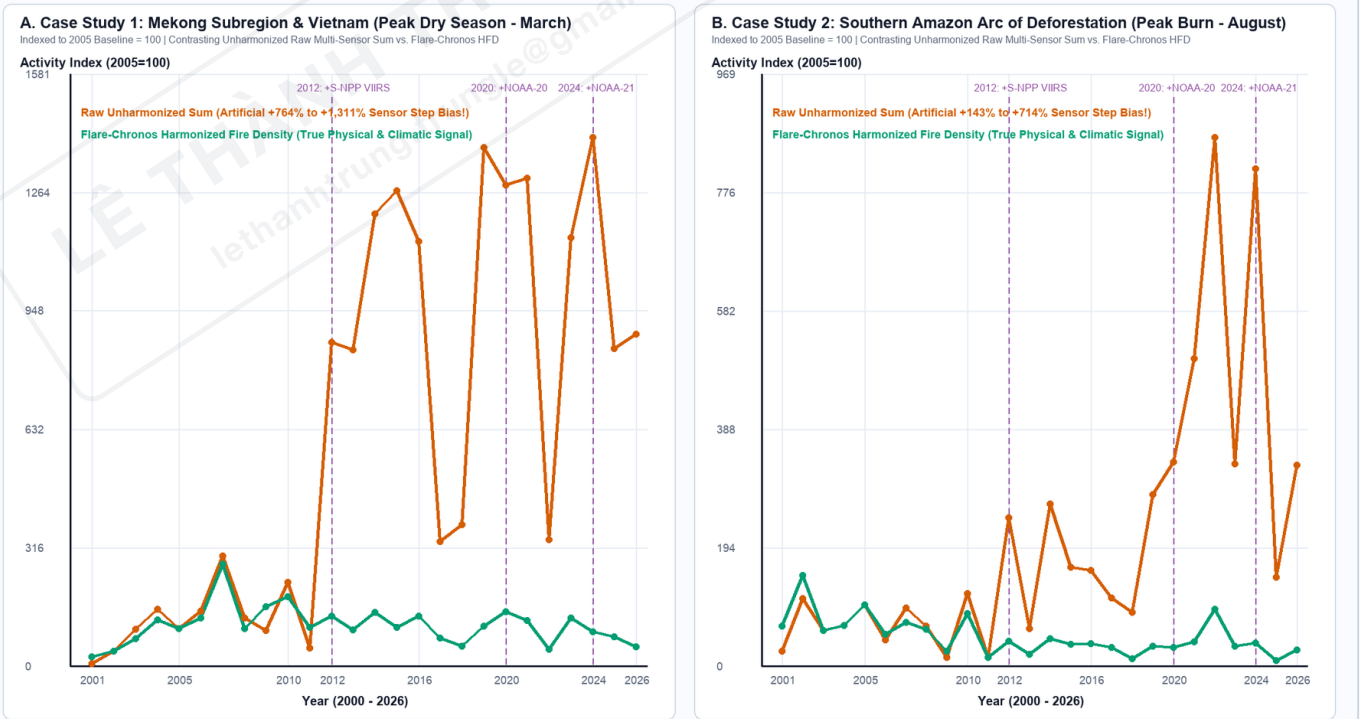


Figure 4 - Regime Shift Elimination

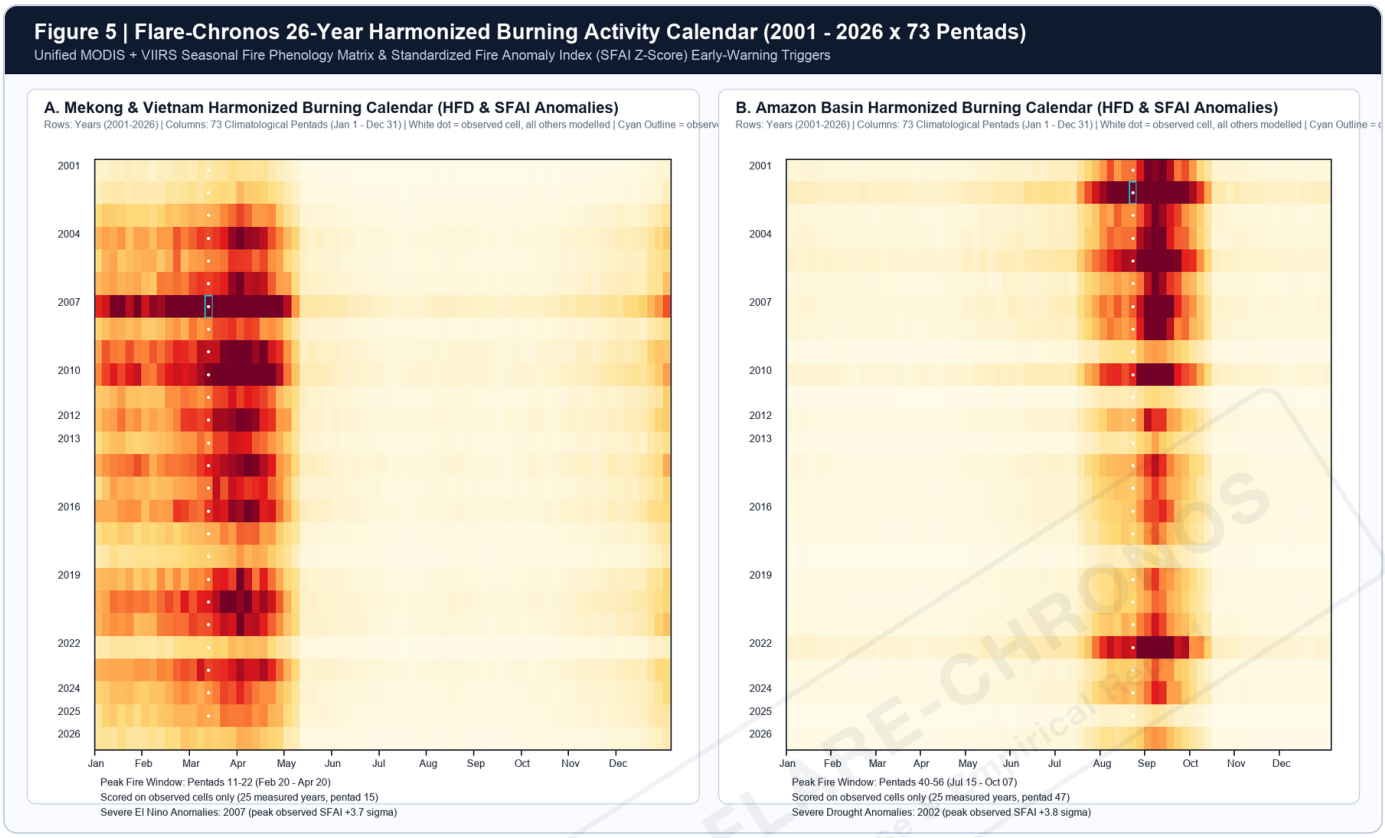
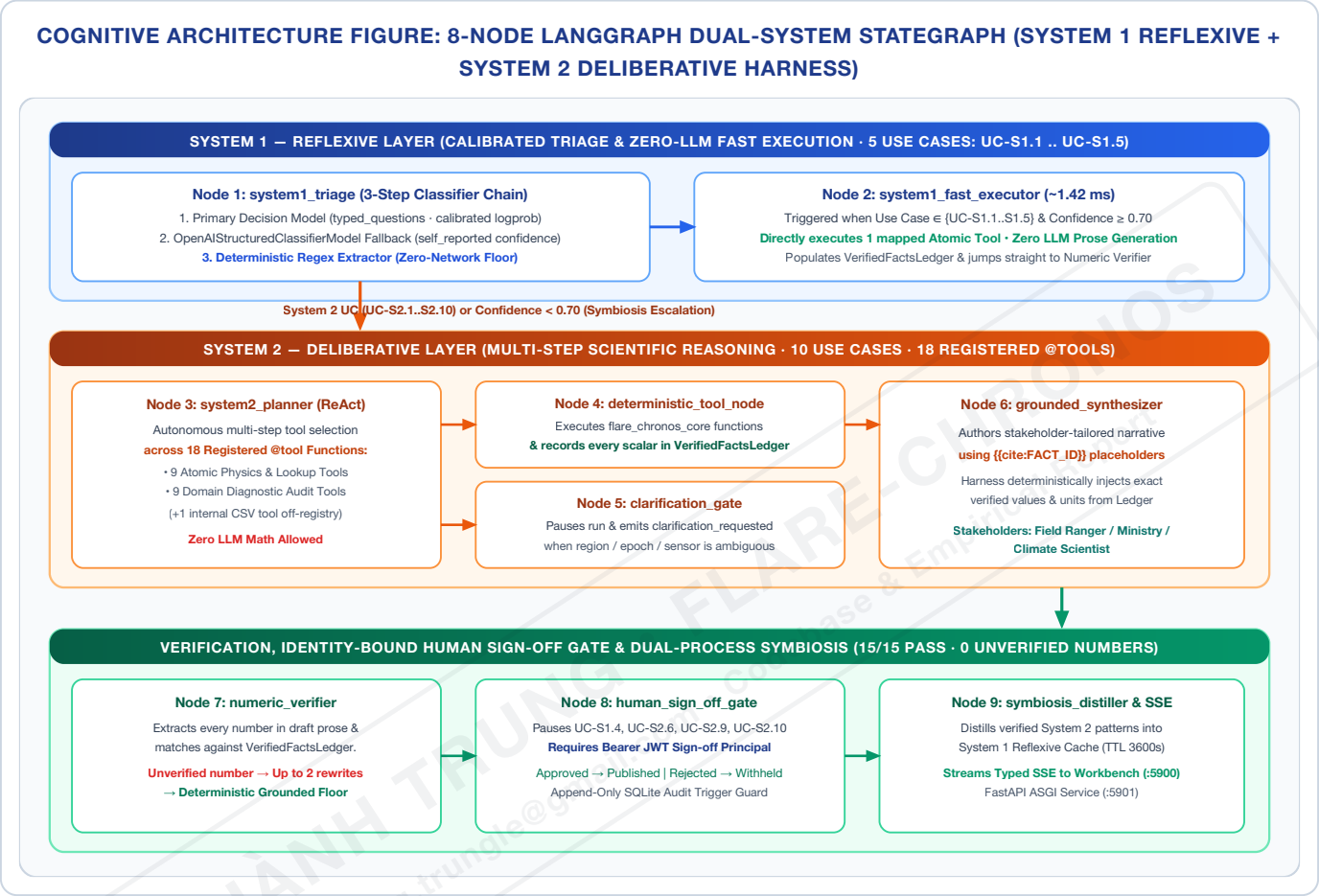


Figure 5 - Harmonized Burning Activity Calendar

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7. Dual-System Cognitive AI Harness (`apps/ai-service` — `15` Use Cases & `18` Registered `@tools`)

7.1. 8-Node LangGraph Dual-System StateGraph Architecture



7.2. Complete Registry of `18` Model-Facing `@tool` Functions (`+ 1` Internal CSV Tool)

1. `9` **Atomic Tools (`ATOMIC_TOOLS` — Accessible to System 1 & System 2)**:
   - `tool_compute_footprint_and_h3_allocation` , `tool_compute_solar_sza_and_daynight` , `tool_evaluate_detection_probability_and_saa` , `tool_calibrate_cross_sensor_frp` , `tool_deduplicate_overpass_episodes` , `tool_lookup_single_epoch_metrics` , `tool_query_seasonal_phenology_12m` , `tool_emit_flash_nrt_alert` , `tool_assemble_stakeholder_briefing_card` .
2. `9` **Domain Diagnostic Tools (`DOMAIN_DIAGNOSTIC_TOOLS` — Accessible Exclusively to System 2)**:
   - `tool_uc1_1_regime_shift_attribution` , `tool_uc1_2_cross_sensor_discrepancy_forensics` , `tool_uc1_3_thermal_source_origin_classification` , `tool_uc2_1_phenological_phase_shift_detection` , `tool_uc2_2_uncertainty_aware_tactical_dispatch` , `tool_uc3_1_constellation_retirement_simulation` , `tool_uc3_3_autonomous_biome_recalibration` , `tool_uc4_1_orbital_drift_and_sza_audit` , `tool_uc4_2_nrt_vs_sp_fidelity_audit` (`18` registered tools total; `tool_harmonize_real_firms_csv_slice` is the `19` th `@tool` kept outside the model registry for raw CSV processing).

7.3. Benchmark Results Across All `15` Use Cases (`5` System 1 + `10` System 2)

- **Deterministic Offline Benchmark (`data/processed/ai_harness_results.json`):** `15 / 15` (`100%`) pass rate, `1.42 ms` mean System 1 latency, `1.61 ms` mean System 2 latency, `0` fallbacks (`logprob` confidence), `0` unverified numbers.

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- **Live Hosted-Model Benchmark (`data/processed/ai_harness_results_live.json`, `gemini-3-flash-preview`)**: `15 / 15` (`100%`) pass rate, `0` unverified numeric claims, `0` unresolved citations, `34,383` total tokens across 15 runs (`15/15` runs gracefully fell back from the offline local `:11434` decision node to Gemini structured classification with `self_reported` confidence provenance).

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8. Interactive Analyst Workbench (`apps/workbench`) & Automated Verification Suite (`tests/`)

8.1. Interactive Analyst Workbench (`apps/workbench`, port `5900`)

Implemented with **Next.js 16 + React 19 + TypeScript**, the Workbench (`apps/workbench`) ships four production routes wired directly to `apps/ai-service` (`:5901`) per ADR-0013 and ADR-0016:

- `/` **(Main Map, 3D Globe, Burning Calendar & Live Briefing Rail)**: Interactive **Three.js 3D Region Globe** with camera fly-to transition into a pre-projected `d3-geo` **1000×730 equal-area SVG map** (`1–64×` pan/zoom), **Uber H3 Resolution 7/8 hexagonal bins**, `26 × 73` Pentad heatmap, and live SSE AI briefing rail.
- `/methodology`: Interactive inspection of the 5 harmonization stages and embedded scientific figures.
- `/review`: Human-in-the-loop sign-off queue reading persisted runs from `GET /api/v1/runs` and submitting identity-bound JWT approval/rejection decisions to `POST /api/v1/runs/{run_id}/resume`.
- `/benchmark`: Interactive 15-Use-Case cognitive evaluation matrix comparing deterministic and live model telemetry.



Interactive 3D Region Globe & Analyst Workbench Preview

8.2. Automated Test Suite (`182` Offline Tests + `6` Live Tests Across `22` Files)

Test Suite Category	Test Files in `tests/`	Test Count & Verification Scope
1. Physics & Harmonization Core	`test_harmonization_suite.py`, `test_core_coverage.py`, `test_sza_and_confidence_boundaries.py`, `test_radiometric_calibrator_edge_cases.py`, `test_episode_clustering_edges.py`, `test_burning_calendar_edges.py`, `test_pipeline_integration.py`	**48 offline tests** — Spherical scan geometry, H3 energy conservation (**$\sum w = 1$**), SZA **$< 85^\circ$**, SAA filtering, EQM calibration, **$\tau = 60$ min** clustering, and SFAI Z-scores.
2. Archive Provenance & Calendar Integrity	`test_raw_data_integrity.py`, `test_epoch_plan.py`, `test_calendar_provenance.py`, `test_documented_claims_consistency.py`, `test_check_docs.py`	**36 offline tests** — SHA-256 verification of all `134` non-empty CSV slices (`1,609,182` rows), `25/26` Measured Epoch provenance, and cross-document claim consistency.
3. Dual-System Cognitive AI Harness & Classifiers	`test_ai_service_harness.py`, `test_openai_structured_classifier.py`, `test_system1_classifier_calibration.py`, `test_endpoint_profiles_and_grounding.py`, `test_ai_service_edge_cases.py`, `test_run_all_usecases_cli.py`, `test_verify_live_Provider.py`	**73 offline tests** — 8-node LangGraph routing, `15` Use Cases, `18` registered `@tools`, `VerifiedFactsLedger`, `NumericGroundingVerifier`, and classifier fallback chains.
4. FastAPI ASGI Server, JWT Auth & SQLite Audit	`test_ai_service_api.py`, `test_review_projection.py`	**25 offline tests** — Bearer JWT authentication, identity-bound human sign-off, append-only SQLite `audit` triggers, and SSE framing.
5. Live Hosted-Model Vertical Slices (`-m live`)	`test_live_tdd_seams.py`	**6 live tests** — End-to-end live LLM classification, multi-tool ReAct deliberation, symbiosis escalation, and real CSV harmonization.
Total Automated Python Test Suite	`22` test modules under `tests/`	`182` **offline tests +** `6` **live tests =** `188` **total tests (** `100%` **passing)**

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